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20AM4T02 : FUNDAMENTALS OF MACHINE LEARNING				

COURSE OBJECTIVES:

1. To understand the need for machine learning and its diverse application.
2. To understand the concepts and algorithms of regression and classification models.
3. To understand graph models and clustering techniques.
4. To understand the ensemble models and its techniques.
5. To understand and evaluate the machine learning models built using available techniques.

COURSE OUTCOMES:

On completion of the course, the student will be able to

CO1: Understand the concepts and types of models in machine learning
CO2: Reduce the dimension of the data set using machine learning techniques
CO3: Implement a classification model for any real scenario
CO4: Implement an unsupervised learning application using clustering techniques
CO5: Implement the ensemble model
CO6: Implement model evaluations for built machine learning model.

UNIT 1:

INTRODUCTION TO MACHINE LEARNING: What is machine learning – Statistics vs. Machine Learning techniques – Supervised learning – Unsupervised learning – Semi-supervised learning – Reinforcement learning - Basics: Probability Theory-Linear Algebra.

UNIT 2:

DECISION THEORY: Statistical Decision Theory: Regression – Classification – Linear regression – Multivariate regression – Logistic regression – Classification predictive modeling – Binary classification – multi-class classification – multi-label classification

UNIT 3:

DIMENSIONALITY REDUCTION: Dimensionality Reduction – Subset Selection – Shrinkage Methods – Linear Discriminant Analysis –Principal Component Analysis – Separating hyper planes.

UNIT4:

PARAMETRIC ESTIMATION: Parameter Estimation – Maximum Likelihood Estimation –Bayesian Parameter Estimation–Support vector Machine – Decision Tree – Naïve Bayes' – Clustering methods – Partitional Clustering-Hierarchical clustering.

UNIT 5:

ENSEMBLE METHODS: Ensemble models – Applications of Ensembling – Types and techniques of Ensembling – Evaluation measures – Hypothesis testing – Cross-validation and Hyper parameter optimization – Bootstrapping and Uncertainties

Text Books:

1. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", 1st Edition, O'Reilly, 2016.
2. Andriy Burkov, "The Hundred-Page Machine Learning Book", 1st Edition, Notion Press, 2019.

Reference Books:

1. Zhi-Hua Zhou, "Ensemble Methods: Foundations and Algorithms", CRC Press, 2012
2. https://machinelearningmastery.com/stacking-ensemble-machine-learning-with-python/
3. https://sebastianraschka.com/blog/2018/model-evaluation-selection-part4.html